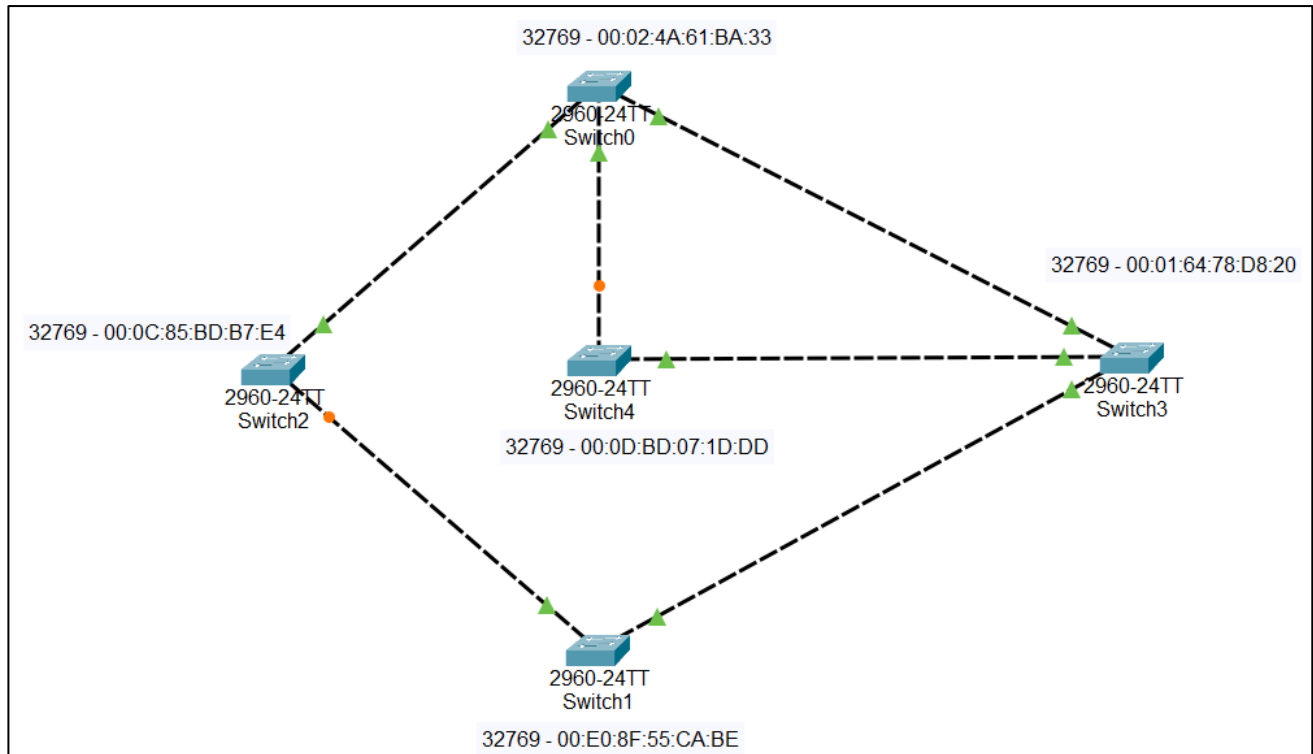




Lab 4 – Spanning Tree Protocol

Consider the following topology formed of 5 switches.



1. Connect the switches between them with crossed wires.
2. Note the MAC addresses of the switch to get it into CLI and use the **show version** command (you can copy the MAC address).

```
64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address       : 00:02:4A:61:BA:33
Motherboard assembly number     : 73-10390-03
Power supply part number        : 341-0097-02
Motherboard serial number       : FOC10093R12
Power supply serial number       : AZS1007032H
Model revision number           : B0
Motherboard revision number      : B0
Model number                    : WS-C2960-24TT-L
System serial number            : FOC1010X104
Top Assembly Part Number        : 800-27221-02
Top Assembly Revision Number    : A0
Version ID                      : V02
CLEI Code Number                : COM3L00BRA
Hardware Board Revision Number  : 0x01
```

3. Note the blocked ports.
4. On each switch, use the **"show spanning-tree"** command and note the role of each of its ports.

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.6478.D820
             Cost        19
             Port        6(FastEthernet0/6)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 se

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     000D.BD07.1DDD
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 se
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/5	Altn	BLK	19	128.5	P2p
Fa0/6	Root	FWD	19	128.6	P2p

- Note it in a label that you put on the figure.
- If the switch is the root switch, the phrase **"This bridge is the root"** will be displayed when running the command **"show spanning-tree"** and all its ports will be **"Designated Port"**.

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.6478.D820
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0001.6478.D820
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/6	Desg	FWD	19	128.6	P2p
Fa0/3	Desg	FWD	19	128.3	P2p
Fa0/4	Desg	FWD	19	128.4	P2p

- Add a switch and connect it to swith0 and Switch1. View port status and note the changes.
 - Note if the root has been changed or not.
- Choose another switch and force it to be the root by changing its priority using the following command:
 - Switch (config) # **spanning-tree vlan 1 priority 4096**
 - Verify that it is become root.

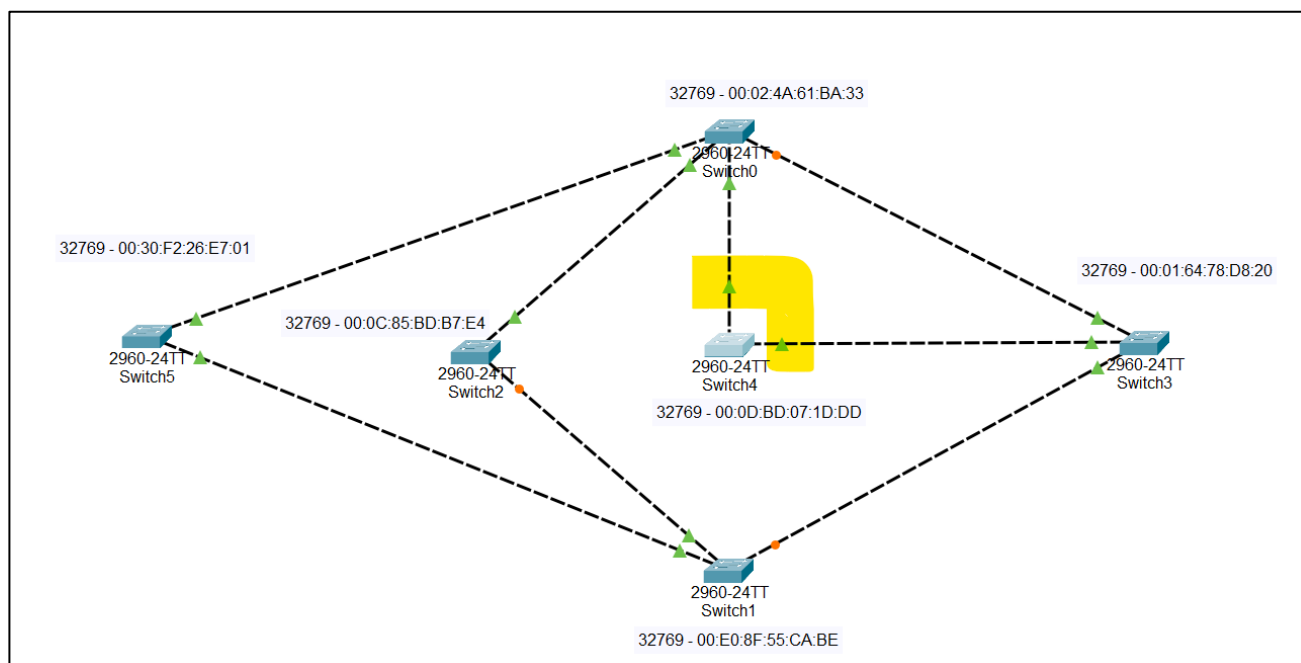
```

Switch#
%SYS-5-CONFIG_I: Configured from console by console
show span
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     000D.BD07.1DDD
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 1

  Bridge ID  Priority    4097 (priority 4096 sys-id-ext 1)
             Address     000D.BD07.1DDD
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 1
             Aging Time 20

Interface                Role Sts Cost          Prio.Nbr Type
-----
Fa0/5                    Desg LRN 19          128.5    P2p
Fa0/6                    Desg FWD 19          128.6    P2p
Switch#

```



c. Try any value as a priority and watch what comes of it.